

Lab 2 – TrashTag Product Specification

Jozsef Kovacs

Old Dominion University

CS411W

Dr. Sarah Hosni

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1 Introduction

This section provides an overview of the TrashTag system and introduces the purpose, scope, and structure of this Software Requirements Specification (SRS). It establishes the context for the system and outlines the key elements necessary to understand the requirements defined in subsequent sections.

1.1 Purpose

The purpose of this SRS is to define the functional and non-functional requirements for the TrashTag prototype system. This document serves as a reference for developers, project stakeholders, and evaluators to ensure a shared understanding of system capabilities and constraints. The SRS establishes a foundation for system design, implementation, validation, and future expansion by clearly describing what the system is expected to do without prescribing specific design or implementation details (IEEE Std 830-1998).

1.2 Scope

TrashTag is a web-based application designed to improve the reporting and coordination of litter cleanup efforts in public and recreational areas. The system enables users to submit geotagged trash reports, including photographic evidence, which are stored in a centralized database and displayed through an interactive mapping interface. The prototype supports core functionality including user authentication, submission of trash reports with images and location data, storage and retrieval of reports, and visualization of reported trash locations on a map. The current prototype does not guarantee cleanup response or coordination by external organizations, does not provide monetary rewards or incentives, and does not function as a law enforcement or emergency reporting system. Real-time notification capabilities are not implemented. The system is intended to demonstrate feasibility and does not represent full production deployment.

1.3 Definitions, Acronyms, and Abbreviations

Geotagging: Associating geographic coordinates with a report.

Trash Report: A user-submitted record containing location, image, and descriptive information.

Prototype: A functional but incomplete implementation.

OpenStreetMap (OSM): Open-source mapping platform.

Authentication: Verification of a user's identity.

1.4 References

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1.5 Overview

This document is organized to provide a structured description of the TrashTag system. Section 1 introduces the purpose, scope, and structure of the SRS. Section 2 provides an overall description of the system, including product perspective, functionality, user characteristics, and constraints. Subsequent sections define detailed system requirements and specifications.

2 Overall Description

The Overall Description section provides a high-level view of the TrashTag system and establishes the context in which the system operates. It describes general factors influencing system

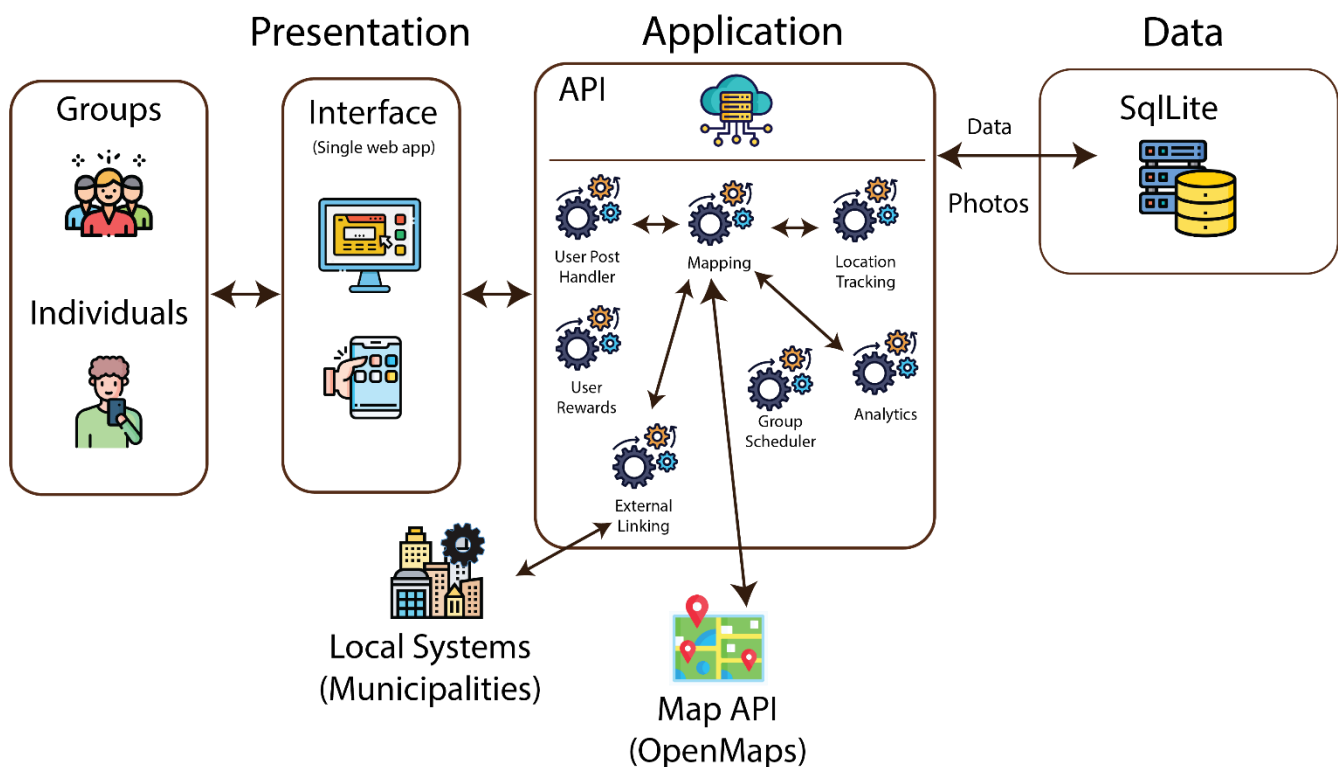
behavior, including its relationship to external components, primary functionality, and user characteristics, consistent with IEEE Std 830-1998.

2.1 Product Perspective

TrashTag is a standalone web application operating within a client-server architecture and integrating OpenStreetMap for geographic visualization. The system is logically separated into presentation, application, and data components. Users interact with the system through web browsers on mobile or desktop devices, and functionality depends on internet connectivity and user-provided or device-captured location data. The system architecture is illustrated in Figure 1.

Figure 1

TrashTag Prototype Major Functional Component Diagram



2.2 Product Functions

The TrashTag prototype provides the following core functions:

Table 1*TrashTag Feature Description and Prototype Implementation*

Function	Description	Prototype Status
User Authentication	Users can create accounts and log in securely	Implemented
Submit Trash Report	Users upload images and associated location data	Implemented
Image Storage	Images are stored with report data in the database	Implemented
Map Visualization	Reports are displayed geographically using Open-StreetMap	Implemented
Report Retrieval	Users can view existing trash reports	Implemented
Cleanup Verification	Users update report status after cleanup	Partially Implemented
Notifications	Users receive alerts about nearby reports	Partially Implemented
Gamification	Ranking or engagement incentives	Partially Implemented

The prototype focuses on validating core reporting and mapping functionality.

2.3 User Characteristics

The system supports general users and administrators. General users have minimal technical expertise and submit/view reports. Administrators manage system integrity, including user and data management.

2.4 Constraints

The system depends on OpenStreetMap and internet connectivity. It does not assume legal liability for user actions and does not enforce compliance with regulations. Location accuracy is not guaranteed.

2.5 Assumptions and Dependencies

Users are assumed to provide accurate data and have capable devices. The system depends on OpenStreetMap, backend storage, and authentication services.